

Marlow Fawn

marlow@mfawn.net | mfawn.net

Education

Karlsruhe Institute of Technology, Karlsruhe, Germany
PhD in Human-Robot Interaction

April 2026 – Ongoing

Tufts University, Medford, MA
M.S. in Computer Science, Human-Robot Interaction; GPA: 3.9

Jun 2023 – Aug 2025

Tufts University, Medford, MA
B.S. in Mechanical Engineering
James P. O’Leary Design Award, Dean’s List

Sep 2015 – May 2019

Research & Experience

MASE Lab, Karlsruhe, Germany *PhD Researcher*

April 2026 – Ongoing

- Integrating contemporary virtue ethics with real robotics hardware to enable context sensitive social interaction.

Tufts Human-Robot Interaction Lab, Medford, MA *Robotics Engineer*

Feb 2023 – August 2025

- Implemented cutting-edge robotics research onto real hardware (Spot, Kinova, UR), including reinforcement learning, realtime vision, and natural language; supervised and mentored researchers in their integrations.
- Created HRI algorithms for learning social norms, handling ASR errors, and multi-agent planning.
- Led a team in the creation of a GUI for dynamic visualization of a robotic cognitive architecture.

Thinking Robots Inc., Boston, MA *Software Engineer*

Dec 2019 – Jan 2023

- Designed and built a mobile robot capable of autonomously connecting to wall outlets to power payloads.
- Developed natural language programming tools enabling non-experts to teach tasks across heterogeneous robots.
- Led multi-robot, multi-human demonstrations in manufacturing and search-and-rescue settings.

Tufts HRI Lab, Medford, MA *Researcher*

2017-2021 (Intermittent)

- Integrated automated planning and novelty handling into a cognitive architecture.

Sustainable Electrochemical Energy Lab, Medford, MA *Researcher*

Sep 2017 – Jun 2018

Superconductivity and Fusion Research Lab, Medford, MA *Researcher*

Sep 2017 – May 2018

Publications

- **Mapping Target-Centered Virtue Ethics onto POMDPs for Value-Transparent Autonomous Systems**
Human State-Aware Robotics Workshop, RO-MAN 2026, *Best Paper Award*
- **Generating Human-Understandable Descriptions of Novel Objects for Verbal Interactions with Edge-Based Robots**
ROBOVIS 2026, *Best Paper Award*
- **Achieving Safe Control Online through Integration of Harmonic Control Lyapunov-Barrier Functions with Unsafe Object-Centric Action Policies**
International Workshop on Formal Methods for Autonomous Systems, iFM 2025
- **On Evaluating LLM Integration into Robotic Architectures**
Transactions on Intelligent Systems and Technology

Skills

- **Robotics & AI:** ROS/ROS2, reinforcement learning, symbolic planning, vision, simulation, LangGraph
- **Programming:** Python, Java, C/C++, JavaScript (React, Firebase), UNIX/Linux
- **Engineering:** CAD, FEA, machining, 3D printing, rapid prototyping